Chinese Journal of Catalysis 37 (2016) 510–516

available at www.sciencedirect.com

ScienceDirect

journal homepage: www.elsevier.com/locate/chnjc

Article

Effects of zinc on Fe-based catalysts during the synthesis of light olefins from the Fischer-Tropsch process

Xinhua Gao [a,b], Jianli Zhang [a,*], Ning Chen [a], Qingxiang Ma [a], Subing Fan [a], Tiansheng Zhao [a,#], Noritatsu Tsubaki [b]

[a] *State Key Laboratory Cultivation Base of Natural Gas Conversion, College of Chemistry & Chemical Engineering, Ningxia University, Yinchuan 750021, Ningxia, China*

[b] *Department of Applied Chemistry, School of Engineering, University of Toyama, Toyama 930-8555, Japan*

ARTICLE INFO

Article history:
Received 12 December 2015
Accepted 15 February 2016
Published 5 April 2016

Keywords:
Zn promoter
Fe-based catalyst
Light olefin
Fischer-Tropsch synthesis
Microwave-hydrothermal method

ABSTRACT

Fe-based catalysts for the production of light olefins via the Fischer-Tropsch synthesis were modified by adding a Zn promoter using both microwave-hydrothermal and impregnation methods. The physicochemical properties of the resulting catalysts were determined by scanning electron microscopy, the Brunauer-Emmett-Teller method, X-ray diffraction, H_2 temperature-programed reduction and X-ray photoelectron spectroscopy. The results demonstrate that the addition of a Zn promoter improves both the light olefin selectivity over the catalyst and the catalyst stability. The catalysts prepared via the impregnation method, which contain greater quantities of surface ZnO, exhibit severe carbon deposition following activity trials. In contrast, those materials synthesized using the microwave-hydrothermal approach show improved dispersion of Zn and Fe phases and decreased carbon deposition, and so exhibit better CO conversion and stability.

1. Introduction

Light olefins (C_2 to C_4) are widely used in the synthesis of plastics, drugs, solvents and cosmetics. In conventional industrial processes, light olefin products are produced through the catalytic cracking of crude oil feedstocks [1,2]. However, because of the growing global demand for bulk chemicals and the ongoing depletion of petroleum reserves, the industrial production of light olefins is threatened by both environmental and economic factors. Recently, the synthesis of light olefins from syngas (CO/H_2) has received significant attention because this approach derives olefins from nonpetroleum carbon resources such as coal, natural gas and biomass [1–3]. In general, two or more stages are required for such syntheses, including the cracking of Fischer-Tropsch (FT)-derived heavy hydrocarbons and the dehydration of FT-derived methanol to light olefins [2,4]. Compared with these multistage reaction processes, a Fischer-Tropsch olefins (FTO) synthesis directly from syngas without intermediate steps would be more energy efficient and economical [1,4].

Fe-based catalysts allow the selective formation of light olefins from CO hydrogenation in reactions that are less expensive and that show low methane and high olefin selectivity, compared with the results obtained from catalysts based on Co, Ru and other metals [1,5–7]. In the case of Fe-based catalysts, there has been much research focused on the optimization of

* Corresponding author. Tel: +86-951-2062393; Fax: +86-951-2062323; E-mail: zhangjl@nxu.edu.cn
# Corresponding author. Tel: +86-951-2062393; Fax: +86-951-2062323; E-mail: zhaots@nxu.edu.cn
This work was supported by the Key Project of Natural Science Foundation of Ningxia (NZ13010) and the National Natural Science Foundation of China (21366025).
DOI: 10.1016/S1872-2067(15)61051-8 | http://www.sciencedirect.com/science/journal/18722067 | Chin. J. Catal., Vol. 37, No. 4, April 2016

support substances such as α-Al_2O_3, SiO_2 or carbon materials [2,4]. However, compared with these supported catalysts, the use of low cost, unsupported bulk catalysts showing competitive olefin selectivity is still of interest [1]. In addition to the development of supported and unsupported catalysts, many groups have focused on examining possible catalytic promoters. Recently, de Jong's group [2] reported that the addition of Na and S produces Fe-based catalysts with high selectivity for light olefins, while Qiao's group [8] showed that K can act as a promoter when used with iron supported on graphene, and enhances the light olefin selectivity. Furthermore, transition metals such as Mn, Zr, V and Cr on Fe-based catalysts are also effective promoters [7–14]. Despite this prior work, there has been little research regarding the selectivity for CO_2 during the promotion of the FTO reaction over Fe-based catalysts.

Our previous work [15] has shown that Fe-Zr catalysts prepared by a microwave-hydrothermal method improve the catalytic activity during CO conversion and also decrease the selectivity for CO_2. However, these catalysts generate a broad spectrum of products, and so it is necessary to further increase their selectivity for light olefins in the C_2 to C_4 range. A Zn promoter has previously been reported [10,12] to increase FTO reaction rates and to enhance the light olefin selectivity over an iron catalyst, although the possibility of severe carbon deposition has been neglected when examining such Zn-promoted catalysts [13,14].

In the present work, we propose a new strategy with regard to the use of Zn promoters to enhance light olefin selectivity and to decrease carbon deposition on Fe-based catalysts. Herein, the effects of a Zn promoter on the structure and catalytic performance of such catalysts is discussed, with the aim of elucidating the function of the Zn promoter.

2. Experimental

2.1. Catalyst preparation

Fe-Zr precursors with an Fe/Zr atomic ratio of 6 were prepared using a microwave-hydrothermal method, as reported previously [15]. In a typical reaction, both $Fe(NO_3)_3{\cdot}9H_2O$ (Tianjin Guangfu Fine Chemical Research Institute, Tianjin, China) and $ZrO(NO_3)_2{\cdot}2H_2O$ (Tianjin Guangfu Fine Chemical Research Institute) were added to a Teflon tube containing an aqueous urea (Tianjin Guangfu Fine Chemical Research Institute) solution, at an (Fe+Zr)/urea molar ratio of 0.5. This mixture was subsequently subjected to microwave-hydrothermal treatment at 2450 MHz and 1.6 MPa over 2 h. The resulting solid product was filtered, washed with deionized water, dried at 120 °C for 12 h and finally calcined at 500 °C for 3 h to obtain the Fe-Zr precursor.

Fe-based catalysts with different ZnO contents were obtained by impregnating the above Fe-Zr precursors with an aqueous solution of $Zn(NO_3)_2{\cdot}6H_2O$ (Tianjin Guangfu Fine Chemical Research Institute). Following this impregnation, the solid samples were dried at 80 °C for 8 h and then calcined at 350 °C for 3 h. Subsequently, 2 wt% K_2CO_3 (Tianjin Guangfu Fine Chemical Research Institute) was added to each solid sample using incipient wetness impregnation. After drying overnight at 120 °C, catalysts were obtained having specific ZnO contents of 0, 10, 20 and 40 wt%, denoted as Zn0, Zn10, Zn20 and Zn40, respectively.

To compare the impregnation of 20 wt% ZnO on Fe-based catalysts, we also employed $Fe(NO_3)_3{\cdot}9H_2O$, $ZrO(NO_3)_2{\cdot}2H_2O$ and $Zn(NO_3)_2{\cdot}6H_2O$ in a one-step microwave-hydrothermal process to prepare an Fe-Zr precursor containing a Zn promoter. In this trial, 2 wt% K_2CO_3 was also employed in the same manner to improve the catalytic performance. This catalyst is denoted as Zn20[M]. All the above catalyst samples were crushed and sieved to obtain particles of sizes 20–40 mesh prior to characterization and activity tests.

2.2. Catalyst characterization

The Brunauer-Emmett-Teller (BET) surface areas of the fresh catalysts were measured at −196 °C using a JW-BK132F N_2 physisorption instrument (Beijing JWGB Sci. & Tech. Co., Ltd.). In these trials, 200-mg samples were held under vacuum at 300 °C for 2 h prior to each measurement. The surface areas were determined using the BET equation based on the single point method. The microscopic morphologies of specimens were obtained by scanning electron microscopy (SEM) with a KYKY-2800B instrument at an accelerating voltage of 25 kV. Powder X-ray diffraction (XRD) patterns of both fresh and used catalysts were acquired with a Rigaku D/MAX-2200PC X-ray diffractometer with Cu K_α radiation (λ = 0.15406 nm) at 40 kV and 30 mA and at a scan rate of 8°/min from 5° to 85° (2θ). H_2 temperature-programmed reduction (H_2-TPR) measurements of fresh catalysts were carried out on a TP-5000 analyzer equipped with a quartz tube reactor. In these measurements, 50-mg catalyst samples were calcined at 500 °C and then pretreated with He (30 mL/min) at 350 °C for 1 h, after which they were cooled to room temperature. Each sample was exposed to a 5% H_2-95% N_2 mixture at a flow rate of 30 mL/min with heating from room temperature to 890 °C at a rate of 10 °C/min. X-ray photoelectron spectroscopy (XPS) data were obtained for both fresh and used samples with a Thermo Scientific ESCALAB 250 spectrometer using an Al K_α X-ray source, with the base pressure of the chamber below 20 nPa. The binding energy (BE) was calibrated relative to that of adventitious carbon, using the C 1*s* peak at 284.8 eV.

2.3. Catalytic reaction

The FTO reaction was carried out in a micro-type fixed-bed reactor system (i.d. = 8 mm; length = 400 mm) with 2 mL catalyst samples. Prior to each reaction, each catalyst sample was pretreated in the syngas flow (H_2/CO = 2) at 1000 h^{-1}, 280 °C and 0.1 MPa for 4 h. The temperature was subsequently increased to 340 °C and pressure was raised to 1.5 MPa to start the reaction. The products were analyzed with an on-line gas chromatograph (GC-9560-I), using a 2-m TDX-01 packed column for C_1 products analysis and a 50-m Al_2O_3 capillary column for C_1–C_5 hydrocarbons products. The liquid products were analyzed on an off-line gas chromatograph (GC-9560-II), with a

2-m GDX-401 packed column and a 30-m SE-30 capillary column for the aqueous and oil phase products, respectively. The mass balance calculations were based on carbon. The CO_2 selectivity was determined using the molar percentage of CO converted to CO_2 out of the total CO converted, and the hydrocarbon distribution was expressed as the mass percentage of desired components out of all hydrocarbons. The catalytic performance of various samples were assessed following reaction trials of up to 200 h.

3. Results and discussion

3.1. Textural properties of the catalyst samples

The textural properties of the various catalyst samples were analyzed by N_2 physisorption. The BET surface areas, pore volumes and average pore sizes determined in this manner are summarized in Table 1. These results indicate that the BET surface areas and the pore volumes decrease with increases in the Zn content, while the average pore sizes increase from Zn0 to Zn20, then decrease from Zn20 to Zn40. Furthermore, compared with the Zn20, the Zn20[M] obtained via a one-step microwave-hydrothermal synthesis has a larger surface area and greater pore volume.

The SEM images of these specimens in Fig. 1 show differences in morphology between the Zn0, Zn20 and Zn20[M]. Interestingly, the Zn20[M] exhibits a greater amount of well dispersed, uniform particles compared with the Zn20. This result indicates that one-step microwave-hydrothermal synthesis improves the catalyst morphology, thus enhancing the efficiency of the Zn promoter.

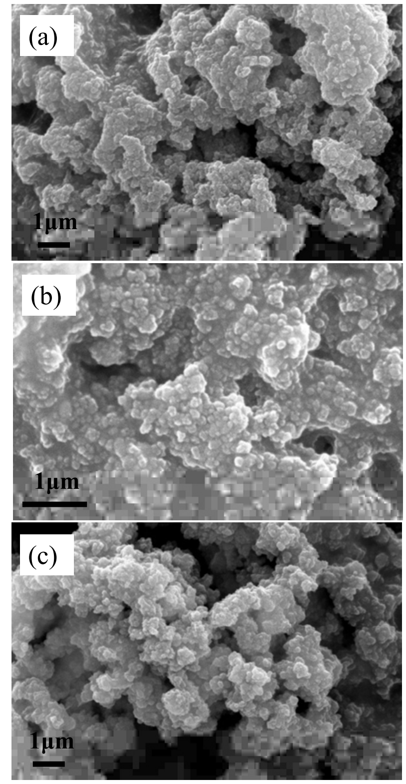


Fig. 1. SEM images of Fe-based catalyst samples. (a) Zn0; (b) Zn20; (c) Zn20[M].

3.2. XRD patterns of the catalyst samples

The XRD patterns of the fresh and used catalysts are presented in Fig. 2. The diffraction peaks at 2θ = 24.2°, 33.3°, 35.7°, 40.9°, 49.6°, 54.1°, 57.6°, 62.6° and 64.1° can be ascribed to rhombohedral hematite (α-Fe_2O_3) [14], while those at 2θ = 30.30°, 35.18°, 50.56°, 60.10°, 63.00° and 74.38° result from ZrO_2 [16]. The peaks attributed to a ZnO phase with a hexagonal wurtzite structure appear at 2θ = 31.77°, 34.42°, 36.26°, 47.54°, 56.60°, 62.86°, 66.86°, 67.94° and 69.06° [17] and evidently increase with increasing Zn content. Interestingly, the Zn20[M] did not generate any ZnO phase peaks, but did appear to produce more intense Fe phase peaks. This suggests that the improved dispersion of the Zn promoter on the Zn20[M] results in less of the catalyst surface being covered compared with the Zn20, such that more active Fe sites are available for the FTO reaction. The α-Fe_2O_3 crystallite sizes on each of the fresh catalysts are quite similar, at 15–18 nm, based on calculations using the Scherrer equation.

Following their use in the FTO reaction, the diffraction peaks of the catalysts decrease and new diffraction peaks attributed to $ZnFe_2O_4$ appear. This may be due to carbon deposition on the catalysts as well as a change in the sample compositions resulting from both reduction and the reaction procedure, respectively [14]. The average $ZnFe_2O_4$ crystallite size of the used catalysts were calculated, and the Zn20[M] crystallites (with a size of 20 nm) were found to be smaller than the Zn20

Table 1
Textural properties of fresh Fe-based catalyst samples.

Sample	BET surface area (m²/g)	Pore volume (cm³/g)	Average pore size (nm)
Zn0	44.5	0.14	12.6
Zn10	37.5	0.12	13.0
Zn20	31.0	0.11	13.9
Zn40	27.5	0.08	11.0
Zn20[M]	43.7	0.13	11.6

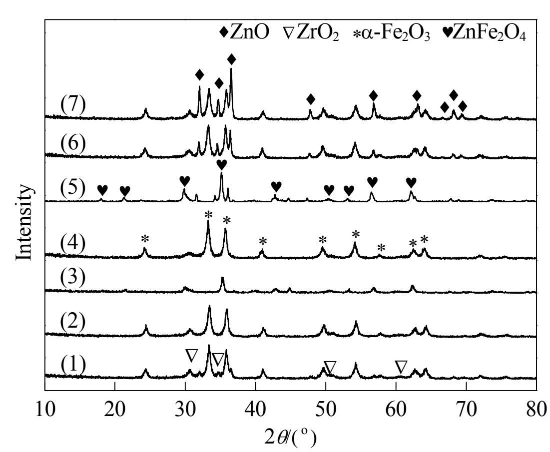


Fig. 2. XRD patterns of the Fe-based catalyst samples. (1) Zn0; (2) Zn10; (3) Zn20[M](used); (4) Zn20[M]; (5) Zn20(used); (6) Zn20; (7) Zn40.

(25 nm) crystallites. It was difficult to identify iron carbides in these materials because of the low diffraction peak intensity in the 2θ range from 43° to 45°. No potassium peaks are detected in the patterns for both the fresh and the used catalysts, possibly due to the low concentration and high dispersion of the element.

3.3. Reduction behavior of the catalyst samples

Fig. 3 shows the H_2-TPR profiles of the Zn0, Zn20 and Zn20[M]. The reduction behavior of the Fe phases can be partitioned into two H_2 consumption temperature ranges. The first peak appears at temperatures below 500 °C while a second, broader peak is found above 500 °C. These are assigned to the α-Fe_2O_3 → Fe_3O_4 and Fe_3O_4 → FeO → Fe transitions, respectively [10]. It is evident that the temperature associated with the α-Fe_2O_3 to Fe_3O_4 reduction is affected by the presence of the Zn promoter. Compared with the Zn-free sample (Zn0), the Zn20 and Zn20[M] undergo the reduction of α-Fe_2O_3 to Fe_3O_4 more readily. In addition, the reduction peaks above 500 °C are also different. This may be the result of the different extents of the two-step reduction (Fe_3O_4 → FeO → Fe) between the catalysts. Assuming ZnO reduction between 400 and 750 °C and sublimation around 600 °C, these differences in the high temperature peaks can be ascribed to variations in the quantities of the Zn promoter as well as to the Fe_3O_4 → FeO → Fe transition over the Zn20 and Zn20[M] [10,14,17].

3.4. XPS characterization of the catalyst samples

The Fe $2p$ XPS spectra of the catalysts are presented in Fig. 4. The Fe $2p_{3/2}$ BE of the Zn0 is approximately 710.9 eV, and a distinct satellite peak about 8 eV higher than the main Fe $2p_{3/2}$ peak can be observed in the spectrum of each fresh sample. The Fe $2p$ BE values and the peak shapes indicate the characteristics of Fe_2O_3 phases on the surfaces of the catalyst samples [11]. The Fe $2p$ peaks of the samples containing Zn are seen to have changed compared with that of the unmodified catalyst (Zn0). These peaks are shifted to lower BE in the vicinity of 710.5 eV for all the fresh and used samples containing the Zn promoter. These data indicate electron transfer from the Zn to the Fe species, an effect that enhances the electron density at the Fe sites and also changes the chemical environment of the Fe. With increasing Zn content, there is no evident change in the BE values. In the case of the used catalysts, the Fe $2p$ signal is greatly attenuated by carbon deposition.

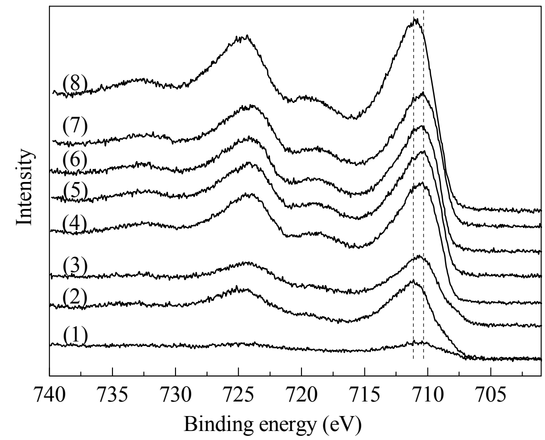


Fig. 4. Fe $2p$ XPS spectra of Fe-based catalyst samples. (1) Zn20(used); (2) Zn0(used); (3) Zn20[M](used); (4) Zn20[M]; (5) Zn40; (6) Zn20; (7) Zn10; (8) Zn0.

To obtain additional information regarding the effects of the Zn promoter on these catalysts, the catalyst samples were further analyzed by XPS, with the results displayed in Fig. 5. The Zn $2p$ peaks of the Zn20 and Zn40 samples are shifted to higher binding energy compared with the Zn10. This indicates that the Zn promoter on these Fe-based catalysts significantly affected the chemical environment of the Fe. Moreover, compared with peak areas in the Zn10, Zn20 and Zn40 spectra, the spectrum of the Zn20[M] shows a smaller peak area. This is attributed to greater dispersion of the Zn promoter and a lower Zn content on the surface of the Zn20[M]. This result is in agreement with the XRD data. Liu et al. [18] have reported that Zn acts as a hydrogenation catalyst and activates hydrogen by heterogeneous

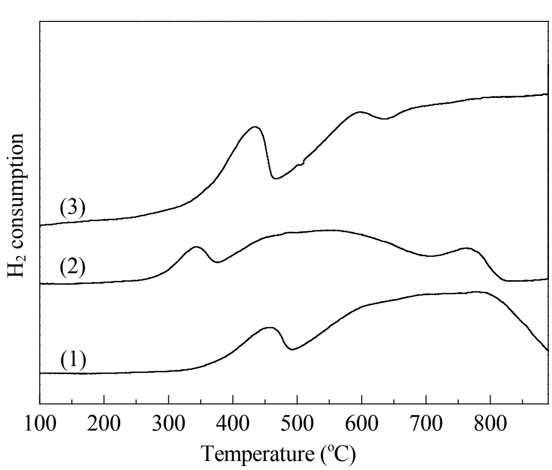


Fig. 3. H_2-TPR profiles of Fe-based catalyst samples. (1) Zn0; (2) Zn20; (3) Zn20[M].

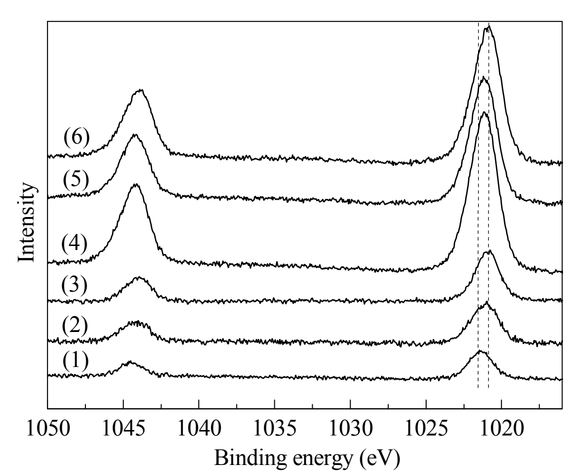


Fig. 5. Zn $2p$ XPS spectra of Fe-based catalyst samples. (1) Zn20(used); (2) Zn20[M](used); (3) Zn20[M]; (4) Zn40; (5) Zn20; (6); Zn10.

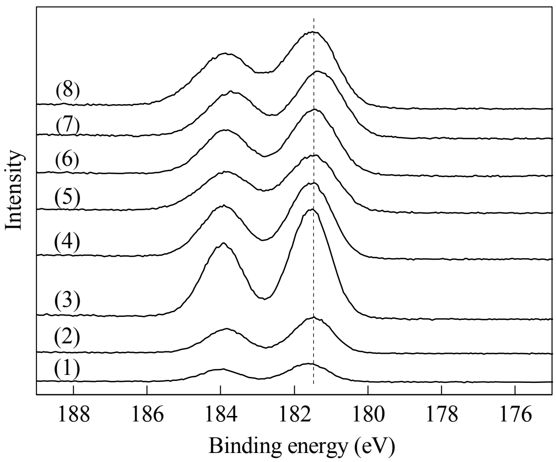


Fig. 6. Zr 3*d* XPS spectra of Fe-based catalyst samples. (1) Zn20(used); (2) Zn0(used); (3) Zn20[M](used); (4) Zn20[M]; (5) Zn40; (6) Zn20; (7) Zn10; (8) Zn0.

splitting, giving rise to ZnH and OH. This process would be expected to provide a large quantity of activated hydrogen to facilitate the reduction of Fe species.

Fig. 6 demonstrates that the Zr 3*d* spectra were largely unchanged following Zn promotion for each of the fresh and used samples, indicating that the chemical environment of the Zr was only minimally affected by the addition of Zn. The surface elemental concentrations of the Zn20[M] and Zn20 before and after the reaction are summarized in Table 2. It is evident that the Zr/Fe and K/Fe molar ratios are very similar over the two fresh samples, while the Zr, Zn and K concentration are enriched over the two samples after the reaction compared with those of the fresh samples. However, the surface Zn content on the Zn20 sample as calculated from the XPS peak areas is 2.2 times that of the Zn20[M] prior to the reaction, further suggesting improved dispersion of Zn over the Zn20[M]. Surprisingly, the ratio of the Zn concentrations on the Zn20 and Zn20[M] changes to 0.5 and the surface carbon concentrations sharply decreases, from 88.2% to 46.0%, following the FTO reaction. These results suggest that the Zn20[M] effectively suppresses carbon deposition.

3.5. Catalytic behaviors of the catalyst samples

The catalytic performance of the samples during the FTO reaction were assessed at 340 °C and 1.5 MPa to test the effects of the Zn promoter. The results show that the selectivity for CO_2 was generally below 25% and that the olefin to paraffin

Table 2
Surface elemental concentrations of Zn20 and Zn20[M] catalyst samples.

Sample	Element content [a] (%)						Atomic ratio [b] (×10)		
	Fe	Zr	Zn	K	O	C	Zr/Fe	Zn/Fe	K/Fe
Zn20	11.0	6.0	4.4	3.7	49.9	25.0	5.5	4.0	3.4
Zn20[M]	15.0	8.2	2.0	5.4	56.3	13.1	5.5	1.3	3.6
Zn20(used)	1.0	1.0	0.7	0.6	8.5	88.2	10.0	7.0	6.3
Zn20[M](used)	4.7	7.4	1.3	5.6	35.0	46.0	15.7	2.8	11.9

[a] Surface element concentrations calculated from XPS data.
[b] Atomic ratios calculated from XPS data.

($C_2^=$–$C_4^=$/C_2^0–C_4^0) ratio over each of the catalysts was greater than 4 (see Table 3). Compared with the sample without Zn, the light olefin selectivity was improved from 38.1% to 40.9% with increasing Zn content in those catalysts made by the impregnation method, although the selectivity for C_5^+ hydrocarbons was decreased over these catalysts. These data suggest that the addition of Zn to the Fe-Zr catalysts improves the distribution of light olefins and favors the production of light hydrocarbons. We propose that the strong interactions between the Fe and Zn suppress the chain growth (and therefore further hydrogenation) of the light olefin products. Conversely, the Zn20[M], having highly dispersed Zn and thus a greater quantity of exposed active Fe sites on its surface, exhibits a high CO conversion of 95.9%. The strong interactions between the Fe and Zn on the Zn20[M] surface were evidently not weakened, because high light olefin selectivity and low C_5^+ selectivity are observed, compared with the sample without Zn.

Furthermore, the Zn20[M] and Zn20 also exhibit excellent stability following 200 h on stream (Fig. 7), such that the olefin selectivity of these catalysts is almost unchanged over this prolonged reaction time. Higher conversion and greater stability were obtained from the Zn20[M] (Fig. 7(a)) compared with the Zn20 (Fig. 7(b)). These results are in agreement with the textural properties of the catalysts, and also with the results obtained regarding optimized chemical environments. Therefore, it appears that the hydrogenation ability of the catalysts is closely related to both the presence of Zn and its dispersion state. The addition of Zn by the impregnation method evidently generates a large amount of ZnO on the surface but also leads to severe carbon deposition, as revealed by XPS studies. The hydrothermal method results in good dispersion of the Zn promoter over the catalyst, decreasing the masking of active Fe sites on the surface and allowing for higher conversion. The reduced carbon deposition on the Zn20[M] can possibly be attributed to enhanced mass transfer and diffusion of the reactants and products resulting from the well dispersed Zn.

Table 3
Catalytic performances of Fe-based catalysts during CO hydrogenation.

Sample	CO conversion (%)	CO_2 selectivity (%)	Product (wt%)				$C_2^=$–$C_4^=$/C_2^0–C_4^0
			CH_4	$C_2^=$–$C_4^=$	C_2^0–C_4^0	CH_5^+	
Zn0	94.2	19.5	20.3	38.1	7.8	33.8	4.9
Zn10	93.5	24.8	23.6	38.5	8.4	29.5	4.6
Zn20	92.9	23.3	21.6	40.8	9.9	27.7	4.1
Zn40	92.8	21.8	24.3	40.1	9.9	25.7	4.1
Zn20[M]	95.9	21.8	21.4	40.9	8.5	29.2	4.8

Reaction conditions: H_2/CO = 2, GHSV = 1000 h^{-1}, p =1.5 MPa, reaction temperature = 340 °C.

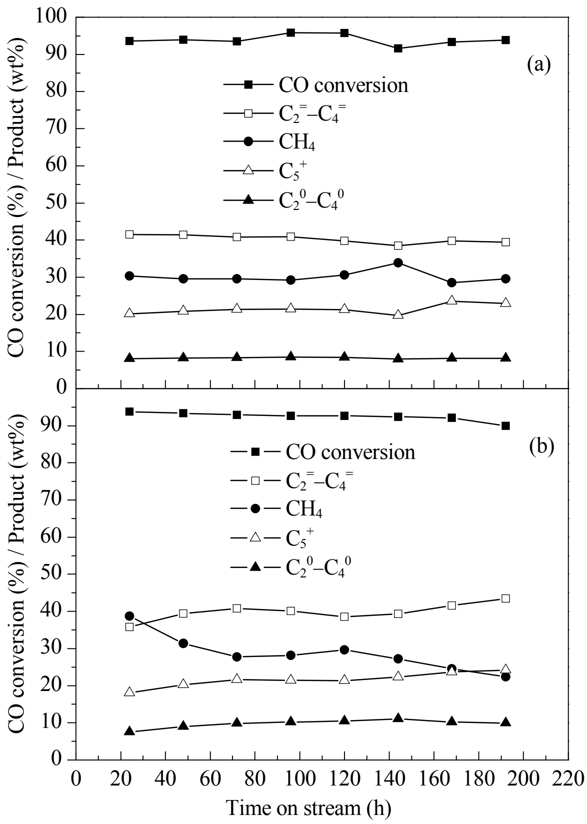


Fig. 7. Stability of (a) Zn20[M] and (b) Zn20 catalysts. Reaction conditions: $H_2/CO = 2$, GHSV = 1000 h^{-1}, p =1.5 MPa, reaction temperature = 340 °C.

4. Conclusions

Zn promoted Fe-based catalysts were found to be favorable for CO hydrogenation to light olefins, exhibiting both high activity and stability. This work demonstrated that the addition of Zn to Fe-based catalysts modifies both the structure and activity of the catalysts, such that lower selectivity for CO_2 and a higher ratio of olefins to paraffins are obtained in all cases. Moreover, the method of Zn addition causes significant differences in the physical and chemical properties of the Fe-based materials. The impregnation method results in a large amount of ZnO being added to the surface but also produces severe carbon deposition. In contrast, the microwave-hydrothermal method significantly improves the dispersion of the Zn and Fe phases and decreases carbon deposition, resulting in better CO conversion and stability.

References

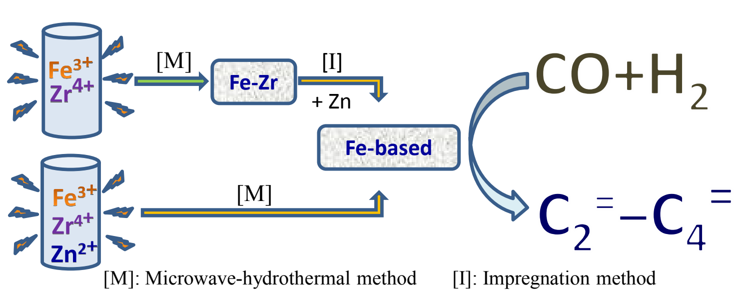
[1] H. M. Torres Galvis, K. P. de Jong, *ACS Catal.*, **2013**, 3, 2130–2149.
[2] H. M. Torres Galvis, A. C. J. Koeken, J. H. Bitter, T. Davidian, M. Ruitenbeek, A. I. Dugulan, K. P. de Jong, *Catal. Today*, **2013**, 215, 95–102.
[3] O. O. James, B. Chowdhury, M. A. Mesubi, S. Maity, *RSC Adv.*, **2012**, 2, 7347–7366.
[4] H. M. Torres Galvis, J. H. Bitter, C. B. Khare, M. Ruitenbeek, A. I. Dugulan, K. P. de Jong, *Science*, **2012**, 335, 835–838.
[5] H. Schulz, *Appl. Catal. A*, **1999**,186, 3–12.
[6] X. Q. Chen, D. H. Deng, X. L. Pan, X. H. Bao, *Chin. J. Catal.*, **2015**, 36, 1631–1637.
[7] J. L. Zhang, L. H. Ma, S. B. Fan, T. S. Zhao, Y. H. Sun, *Fuel*, **2013**, 109, 116–123.
[8] Y. Cheng, J. Lin, K. Xu, H. Wang, X. Y. Yao, Y. Pei, S. R. Yan, M. H. Qiao, B. N. Zong, *ACS Catal.*, **2016**, 6, 389–399.
[9] W. S. Ning, S. B. Yang, H. X. Chen, M. Yamada, *Catal. Commun.*, **2013**, 39, 74–77.
[10] O. O. James, B. Chowdhury, S. Maity, *J. Chem. Sci.*, **2013**, 125, 679–686.
[11] G. C.Wang, K. Zhang, P. Liu, H. T. Hui, Y. S. Tan, *J. Ind. Eng. Chem.*, **2013**, 19, 961–965.
[12] H. L. Wang, Y. Yang, J. Xu, H. Wang, M. Y. Ding, Y. W. Li, *J. Mol. Catal.*, **2010**, 326, 29–40.

Graphical Abstract

Chin. J. Catal., 2016, 37: 510–516 doi: 10.1016/S1872-2067(15)61051-8

Effects of zinc on Fe-based catalysts during the synthesis of light olefins from the Fischer-Tropsch process

Xinhua Gao, Jianli Zhang *, Ning Chen, Qingxiang Ma, Subing Fan, Tiansheng Zhao *, Noritatsu Tsubaki
Ningxia University, China; University of Toyama, Japan



Fe-based Fischer-Tropsch catalysts for light olefins synthesis were modified by adding a Zn promoter using either microwave-hydrothermal or impregnation methods. The catalysts prepared by the microwave-hydrothermal technique exhibited high catalytic activity and good stability with significantly lower carbon deposition.

[13] S. L. Soled, E. Iglesia, S. Miseo, B. A. DeRites, R. A. Fiato, *Top. Catal.*, **1995**, 2, 193–205.
[14] S. Z. Li, A. W. Li, S. Krishnamoorthy, E. Iglesia, *Catal. Lett.*, **2001**, 77, 197–205.
[15] X. H. Gao, Y. L. Wang, S. P. Lu, J. L. Zhang, S. B. Fan, T. S. Zhao. *J. Fuel Chem. Technol.*, **2014**, 42, 219–224.
[16] F. Bondioli, A. M. Ferrari, C. Leonelli, C. Siligardi, G. C. Pellacani, *J. Am. Ceram. Soc.*, **2001**, 84, 2728–2730.
[17] O. W. Perez-Lopez, A. C. Farias, N. R. Marcilio, J. M. C. Bueno, *Mater. Res. Bull.*, **2005**, 40, 2089–2099.
[18] X. M. Liu, G. Q. Lu, Z. F. Yan, J. Beltramini, *Ind. Eng. Chem. Res.*, **2003**, 42, 6518–6530.